ADAPTIVE LEARNING FOR NEURODIVERGENT STUDENTS

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PROBLEM STATEMENT

1 in 5 students is neurodivergent. Curricula are built for the other 4.

The Reality:

- 1 in 5 children is neurodivergent - affecting how they learn, focus, and process information, not their intelligence (EdWeek, 2024)
- In the US alone, 7.5 million students qualify for special education under IDEA - and that's only those formally identified (National Center for Education Statistics, 2022–23)
- Neurodivergent students have the lowest graduation rate of any disability group - 58%, compared to 79% for students without disabilities (UC Academic Senate / UC Davis Aggie Neurodiversity Committee)
- 65% of neurodivergent students in higher education never disclose their condition, which means they receive no accommodations at all (National Center for Education Statistics)

The Cost: These students aren't falling behind because they can't learn. They're falling behind because no one built learning for them.

The curriculum isn't broken. It's just built for someone else.

SOLUTION

We Transform Any Curriculum. Instantly. For Every Brain.

How It Works:

- Teacher uploads any file: a textbook chapter, a syllabus, a lesson plan.
- Our system reads the content and rebuilds it into the format that matches how each student learns:
    - ADHD: short gamified challenge with a clear goal and progress bar.
    - Dyslexia: audio-first with dyslexic-friendly font and word-by-word highlighting.
    - ASD: structured, predictable layout with no visual noise or surprises.
- A short onboarding quiz sets the student's profile - educators can adjust it anytime.
- As students use it, the system picks up on what's working: longer sessions, fewer retries, and quietly shifts the format over time expand on this.

THE PIPELINE

Stage 1: Ingest

- Educator uploads any file: PDF textbook, DOCX syllabus, PowerPoint lesson plan
- OCR and NLP break content into curriculum atoms - the smallest unit of a concept
- Each atom is tagged: subject, grade level, learning objective, Bloom's taxonomy level

Stage 2: Transform

- The student's cognitive profile - a weighted blend of ADHD, Dyslexia, and ASD traits - is matched to the content atom
- An LLM rewrites it into the right format: a gamified quest, an audio-first script, or a structured low-stimulation layout — or a mix of all three for blended profiles
- Every generated output sits in an educator review queue before it reaches a single student

Stage 3: Deliver and Adapt

- Student receives their personalised version inside a familiar portal - no new app to install
- System tracks passive signals: session length, retry count, exit point
- Profile auto-recalibrates weekly - no teacher action needed

TECH STACK

Layer	Technology
Document Ingestion	Apache Tika, Tesseract OCR, spaCy
AI Transformation	GPT-4o via prompt routing
Backend	FastAPI, PostgreSQL, Redis
Frontend	React PWA
Infrastructure	Kubernetes, Terraform
LMS Integration	LTI 1.3 - Google Classroom, Canvas